

Unit 13, Lesson 1: Simple Interest



Benchmarks

MA.912.FL.3.2 Solve real-world problems involving simple, compound, and continuously compounded interest.

MA.912.FL.3.4 Explain the relationship between simple interest and linear growth. Explain the relationship between compound interest and exponential growth, and the relationship between continuously compounded interest and exponential growth.

MA.912.F.1.8 Determine whether a linear, quadratic, or exponential function best models a given real-world situation.



Learning Targets

- ☐ I can explain the relationship between simple interest and linear growth.
- ☐ I can solve real-world problems involving simple interest.



13.1.1 Warm-Up: Earth's Surface Speed

The surface of the Earth at the equator moves at a speed of 460 meters per second.

1. Based on your understanding of meters and miles, do you think converting meters per second to miles per second will result in a larger or smaller number?
2. Use the conversion 1 mile = 1609.34 meters to change 460 meters per second to miles per second.
3. Use the conversion 1 hour = 3600 seconds to determine the speed of the surface of the Earth in miles per hour.



13.1.2 Guided Instruction: Simple Interest

When money is borrowed or saved, interest is typically accrued. For example, if money is borrowed, the interest accrued is owed in addition to the amount borrowed. If money is saved in an interest-bearing account, the interest accrued is saved in addition to the funds deposited into the account.

Simple interest is calculated by using three quantities — the principal, which could be an amount borrowed or an amount invested; the rate of interest; and the time period it will cover.

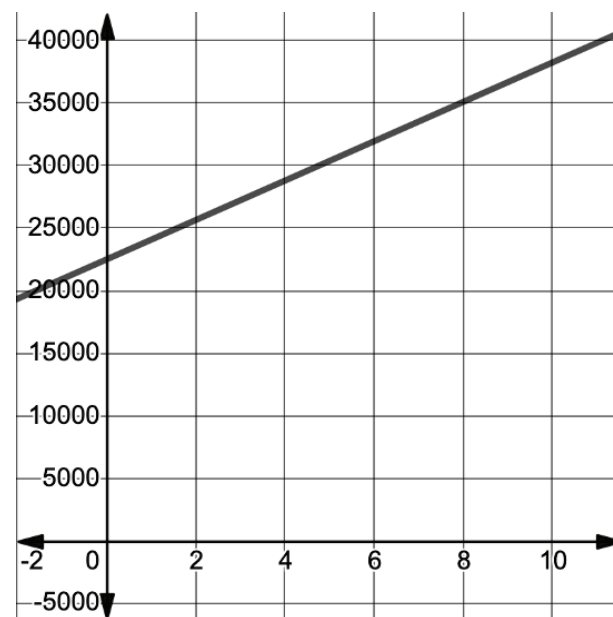
Student loans and mortgages typically use simple interest, whereas savings accounts and credit card debt typically use compound interest.



Simple interest is a method of computing interest. Interest is computed from the (original) principal alone, no matter how much money has accrued so far.

The formula $A = P(1 + rt)$ is used to calculate the final amount owed or saved, where A = final amount, P = principal or original amount, t = number of years, and r = rate of interest per year.

1. Bianca just graduated from college and is ready to buy her first car. She borrows \$22,500 from a bank to make the purchase. The loan she gets has an annual interest rate of 7 percent.



- a. What do you notice about the graph?
- b. Place a point on the graph that represents the final amount of the loan if Bianca pays the loan off in 4 years.
- c. Use the graph to determine how much Bianca will pay for the car if she pays it off in 8 years.

- d. The equation $A = 22,500(1 + 0.07t)$ models Bianca's loan. Rewrite the equation in slope-intercept form.

- e. Work with your partner to complete the table.

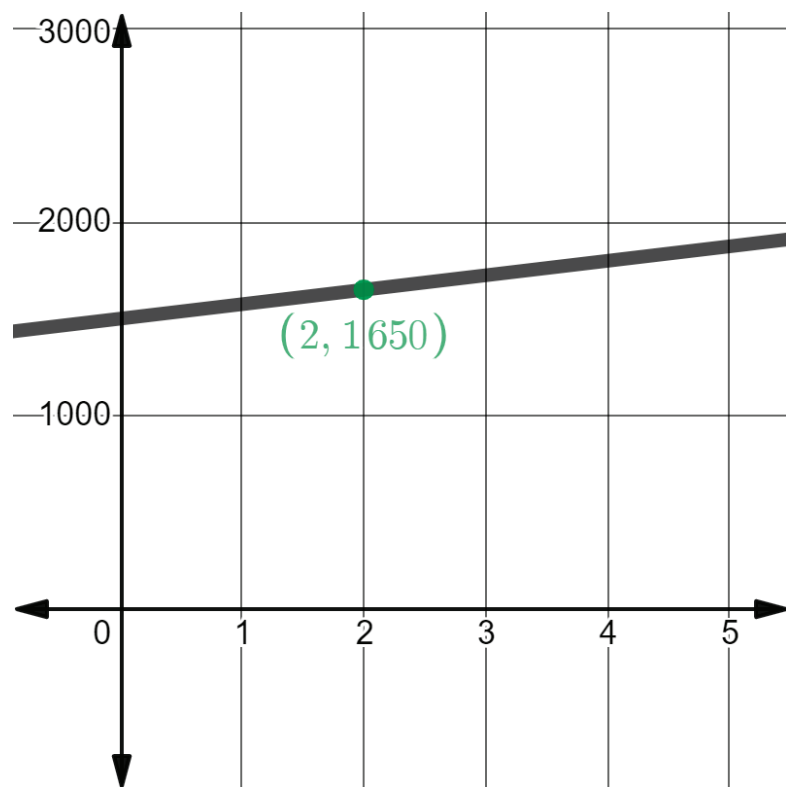
Key Feature	Value	Interpretation in Context
y-intercept		
slope		

- f. Bianca only wants to pay a total of \$27,500 for the car, including interest on her loan. Use the equation $A = 22,500(1 + 0.07t)$ to determine how long the loan can be to ensure she only pays \$27,500. Round to the nearest hundredth.

A **certificate of deposit** (CD) is a type of savings account offered by banks and credit unions that provides a higher interest rate in exchange for the customer agreeing to leave a lump-sum deposit untouched for a predetermined period of time.

2. In 2001, Logan's parents had \$1,500 to invest. They chose a certificate of deposit that offered a 5% annual percentage rate (APR).
- a. Work with your partner to write an equation that models his parents' deposit using the formula $A = P(1 + rt)$, where A = final amount, P = principal or original amount, t = number of years, and r = rate of interest per year.

- b. The graph shows the equation modeling Logan's parents' investment and the point $(2, 1650)$.



What does the point $(2, 1650)$ represent?

The time frames for certificates of deposit are often given in months.

- c. Discuss with your partner if an 18-month CD would have earned less or more money than the point $(2, 1650)$ for Logan's parents. Explain how you determined your answer.
- d. For what length of time, in months, would his parents need a CD if their goal was to have \$1,750?

- e. Complete the table for simple interest relationships.

Key Feature	Description
End Behavior	As $x \rightarrow \infty$, $y \rightarrow$
Increasing/Decreasing	As x increases, $y \rightarrow$

- f. Discuss with your partner how the key features of end behavior and intervals where the function is increasing/decreasing help to explain the relationship between simple interest and linear growth. Summarize your discussion.

- g. What other key feature is useful to explain that simple interest is modeled by linear growth?



13.1.3 Guided Practice: Calculating Simple Interest

1. Randall is a pro fisherman. He bought a new boat for \$77,500 using a loan with a simple interest rate of 3.75%.

- a. Write an equation that models Randall's loan.

- b. Complete the table.

Key Feature	Value	Interpretation in Context
y-intercept		
slope		

- c. What is the total Randall will have to pay if he pays the loan off in 4 years?

- d. How many years did it take Randall to pay off the loan if the total he paid was \$100,750?



13.1.4 Your Turn!

- When Valentina was born, her grandmother put \$2,500 into a savings account for her. The savings account had a 2.25% interest rate. Each year on her birthday, her grandmother told her how much money was in the account.

a. Complete the table. Show your work.

Valentina's Age	Total Amount
7	
18	

- How old will Valentina be when the account has reached \$4,000?



13.1.5 Wrap-Up: Reflection

- Take a **SELFIE** by checking the box that best matches what you did today.

Showed my work	☐	☐
Explained my answers	☐	☐
Lots of math vocabulary used	☐	☐
Found solutions	☐	☐
I persevered through the problem	☐	☐
Eliminated careless errors	☐	☐